Middle Factor of the Eigenvalue Rosetta Stone: $(1+q)^{\lfloor n/2 \rfloor}$ Divisibility of $\mathrm{tr}_{V_\lambda}(\Pi_q)$

Verification at $n = 3, 4, 5$; partial factorization argument

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Abstract

Let $\Pi_q = \prod_{j=1}^{\binom{n}{2}}(1+T_{i_j})$ be the staircase Bott–Samelson element of the Iwahori–Hecke algebra $H_q(S_n)$ with $T_i^2 = (q-1)T_i + q$, and let $H = \langle s_1, s_3, s_5, \ldots\rangle \cong (S_2)^{\lfloor n/2 \rfloor}$ with Poincaré polynomial $(1+q)^{\lfloor n/2 \rfloor}$. We verify the middle factor of the *Eigenvalue Rosetta Stone* conjecture $e_\lambda(q) = q^{n(\lambda)}(1+q)^{\lfloor n/2 \rfloor} p_\lambda(q)$ at $n = 3, 4, 5$: for every irreducible S_n-representation V_λ with $V_\lambda^H \neq 0$, the trace polynomial $\mathrm{tr}_{V_\lambda}(\Pi_q) \in \mathbb{Z}[q]$ is divisible by $(1+q)^{\lfloor n/2 \rfloor}$ in $\mathbb{Z}[q]$, and each nonzero eigenvalue of $\Pi_q|_{V_\lambda}$ (in $\overline{\mathbb{Q}(q)}$) is divisible by $(1+q)^{\lfloor n/2 \rfloor}$.

We also establish a partial algebraic identity: at $n = 4, 5$ the staircase factorizes as $\Pi_q = A \cdot P_H \cdot B$ where $P_H = \prod_{i \text{ odd}}(1+T_i)$ is the H-symmetrizer. This puts $(1+q)^{\lfloor n/2 \rfloor} = (1+q)^{\mathrm{rank}(P_H^2 \to P_H)}$ into the trace via $P_H^2 = (1+q)^{\lfloor n/2 \rfloor} P_H$, *if* integrality of the quotient trace can be shown. We document honestly where the argument passes and where it stalls.

1 Setup

Let $H_q(S_n)$ be the Iwahori–Hecke algebra of S_n with generators $T_1, \ldots, T_{n-1}$ and relations

$$T_i^2 = (q-1)T_i + q, \quad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \quad T_iT_j = T_jT_i \ (|i-j| \geq 2).$$

Fix the *staircase reduced word* for $w_0 \in S_n$:

$$\mathbf{i} = (s_1)(s_2s_1)(s_3s_2s_1)\cdots(s_{n-1}\cdots s_1), \qquad \ell(\mathbf{i}) = \tbinom{n}{2}.$$

Define

$$\Pi_q \;=\; \prod_{j=1}^{\binom{n}{2}}(1+T_{i_j}) \;\in\; H_q(S_n), \qquad P_H \;=\; \prod_{i \in \{1,3,5,\ldots\} \cap [1,n-1]} (1+T_i). \tag{1}$$

Write $s = \lfloor n/2 \rfloor$, so $H = (S_2)^s$ is the parabolic with odd generators and $|H| = 2^s$, Poincaré polynomial $\sum_{w \in H} q^{\ell(w)} = (1+q)^s$.

Two facts used throughout:

Lemma 1. *$(1+T_i)^2 = (1+q)(1+T_i)$; hence $P_H^2 = (1+q)^s P_H$.*

Proof. $(1+T_i)^2 = 1 + 2T_i + T_i^2 = 1 + 2T_i + (q-1)T_i + q = (1+q)(1+T_i)$. Since the odd generators pairwise commute, $P_H = \prod_{i \text{ odd}}(1+T_i)$ has $P_H^2 = \prod_i (1+T_i)^2 = (1+q)^s P_H$. □

Lemma 2. *For every irrep V_λ, $\Pi_q \cdot V_\lambda \subseteq V_\lambda^H$; in fact $\mathrm{im}\,\Pi_q|_{V_\lambda} = V_\lambda^H$ when the latter is nonzero. ("H-invariant theorem," proved 2026-04-17.)*

2 Main theorem

Theorem 3 (Middle factor, verified at $n \le 5$)**.** *For $n \in \{3,4,5\}$ and every partition $\lambda \vdash n$ with $V_\lambda^H \neq 0$:*

(a) ***Trace divisibility.*** $\mathrm{tr}_{V_\lambda}(\Pi_q)$ *is divisible by $(1+q)^s$ in $\mathbb{Z}[q]$, where $s = \lfloor n/2 \rfloor$.*

(b) ***Per-eigenvalue divisibility.*** *Each of the $\dim V_\lambda^H$ nonzero eigenvalues of $\Pi_q|_{V_\lambda}$, as an element of $\overline{\mathbb{Q}(q)}$, is divisible by $(1+q)^s$ in the sense that its quotient by $(1+q)^s$ is integral over $\mathbb{Z}[q]$.*

Proof by direct computation. We use Young's seminormal form in the Hoefsmit/Ram convention. For each standard Young tableau T of shape λ label pairs (T, s_iT) with axial distance $d = \mathrm{ct}(T[i]) - \mathrm{ct}(T[i+1])$ and set $a(d) = (q-1)/(1-q^d)$. On the 2-dimensional block spanned by (v_T, v_{s_iT}) with T ordered before s_iT:

$$T_i v_T = a(d)\, v_T + v_{s_iT}, \qquad T_i v_{s_iT} = \big(a(d)a(-d) + q\big) v_T + a(-d)\, v_{s_iT}.$$

Same-row cells give $T_i v_T = q v_T$; same-column cells give $T_i v_T = -v_T$. We verified for each (λ, n) below that the matrices satisfy $T_i^2 = (q-1)T_i + q$, the braid relation, and commutativity for $|i-j| \ge 2$. Then $\Pi_q|_{V_\lambda}$ is computed by matrix product along the staircase word, and $\mathrm{tr}_{V_\lambda}(\Pi_q)$ is extracted as a $\mathbb{Q}(q)$-expression and checked for polynomial division by $(1+q)^s$.

The results are exhibited in Table 1. Every row with $V_\lambda^H \neq 0$ has $\mathrm{tr}_{V_\lambda}(\Pi_q) \in (1+q)^s\,\mathbb{Z}[q]$; the explicit quotient $\mathrm{tr}_{V_\lambda}(\Pi_q)/(1+q)^s$ is shown.

For (b): when $\dim V_\lambda^H = 1$, the unique nonzero eigenvalue equals the trace, so (a) already implies (b). When $\dim V_\lambda^H = 2$ (occurs only for $\lambda = (4,1), (3,2)$ at $n = 5$), the two nonzero eigenvalues α_1, α_2 satisfy $x^2 - Sx + P = 0$ with $S = \alpha_1 + \alpha_2$ the trace and $P = \alpha_1\alpha_2$ the subleading coefficient of the characteristic polynomial. Substituting $y = x/(1+q)^s$ gives $y^2 - \big(S/(1+q)^s\big)y + \big(P/(1+q)^{2s}\big) = 0$, which has coefficients in $\mathbb{Z}[q]$ iff $(1+q)^s \mid S$ and $(1+q)^{2s} \mid P$. Both are verified:

λ, n	$S = \mathrm{tr}\ \Pi_q$	$P = \alpha_1\alpha_2$
$(4,1), n=5$	$q(1+q)^4(q^4+7q^3+17q^2+7q+1)$	$q^3(1+q)^{12}(q^2+4q+1)$
$(3,2), n=5$	$q^2(1+q)^2(q^4+9q^3+19q^2+9q+1)$	$q^5(1+q)^8(q^2+4q+1)$

In both cases $(1+q)^s$ divides S (with $s = 2$, divides to higher order in fact) and $(1+q)^{2s}$ divides P. □

3 Partial algebraic argument via $\Pi_q = A \cdot P_H \cdot B$

For $n = 4, 5$ the staircase word begins with four factors that align with P_H after a single commutation:

Lemma 4. *For $n = 4$: $\Pi_q = A\, P_H\, B$ with*

$$A = (1+T_1)(1+T_2), \quad P_H = (1+T_1)(1+T_3), \quad B = (1+T_2)(1+T_1).$$

For $n = 5$:

$$A = (1+T_1)(1+T_2), \quad P_H = (1+T_1)(1+T_3), \quad B = (1+T_2)(1+T_1)(1+T_4)(1+T_3)(1+T_2)(1+T_1).$$

Proof. The staircase word for $n = 4$ is $s_1 \cdot (s_2 s_1) \cdot (s_3 s_2 s_1)$ of length 6. Read left-to-right: $(1 + T_1)(1 + T_2)\,(1 + T_1)(1 + T_3)\,(1 + T_2)(1 + T_1)$. Positions $3, 4$ give generators s_1, s_3 which commute; their product is precisely P_H. Identification of A and B by inspection. The $n = 5$ case is the same with B extended by the first four factors of the $s_4 \cdots s_1$ block.

Verified computationally in the length-lex regular representation of $H_q(S_4)$ (24×24) and $H_q(S_5)$ (120×120): every entry of $\Pi_q - AP_HB$ is identically zero. □

Corollary 5 (Cyclic-trace reformulation)**.** *For $n = 4, 5$ and every representation V of $H_q(S_n)$,*

$$\mathrm{tr}_V(\Pi_q) = \mathrm{tr}_V(A\,P_H\,B) = \mathrm{tr}_V(P_H \cdot Y), \qquad Y := BA \in H_q(S_n).$$

Proof. Cyclic property of trace. □

3.1 What would close the proof algebraically

Set $e_H := P_H/(1+q)^s$. On any $H_q(S_n)$-module V over $\mathbb{Q}(q)$, e_H is an idempotent (Lemma 1) with $\mathrm{im}\, e_H = V^H$. Corollary 5 combined with Lemma 1 gives

$$\mathrm{tr}_{V_\lambda}(\Pi_q) \;=\; (1+q)^s\ \mathrm{tr}_{V_\lambda}(e_H \cdot Y) \;=\; (1+q)^s\ \mathrm{tr}_{V_\lambda^H}(Y|_{V_\lambda^H}), \tag{2}$$

where the last equality uses that e_H is the projector onto $V_\lambda^H \subseteq V_\lambda$ and cyclic trace of a projector.

Proposition 6 (Trace divisibility, assuming integrality)**.** *If $\mathrm{tr}_{V_\lambda^H}\big(Y|_{V_\lambda^H}\big) \in \mathbb{Z}[q]$ then $(1+q)^s \mid \mathrm{tr}_{V_\lambda}(\Pi_q)$ in $\mathbb{Z}[q]$.*

Proof. Direct from (2). □

The *a priori* statement is only that this trace is an element of $\mathbb{Q}(q)$. Integrality (no $(1+q)$ in the denominator) is the missing step.

3.2 Where the naive strengthening fails

One is tempted to argue: $P_H \cdot H_q(S_n) \cdot P_H \subseteq (1+q)^s H_q(S_n)$, whence $P_H Y P_H$ is divisible in $H_q(S_n)$ and its trace inherits the divisibility. This *fails*:

Proposition 7 (Counterexample at $n = 4$)**.** *The ideal $P_H H_q(S_n) P_H \subseteq H_q(S_4)$ is not contained in $(1+q)^2 H_q(S_4)$. Specifically, for the double coset $Hs_2H \subseteq S_4$ one computes $P_H \cdot T_{s_2} \cdot P_H \notin (1+q)^2 H_q(S_4)$; some coefficients in the T-basis are in $\mathbb{Z}[q] \setminus (1+q)^2\mathbb{Z}[q]$.*

Proof. Direct computation in the left-regular representation of $H_q(S_4)$. Details in the verification script `check_PH_yP_H_v2.py`; the minimum k such that $(1+q)^k$ divides every T-basis coefficient of $P_H T_{s_2} P_H$ is $k = 0$, not $k = 2$. □

Remark 8. The double-coset decomposition $S_n = \coprod_d H\sigma_d H$ partitions $P_H H_q(S_n) P_H = \sum_d P_H T_{\sigma_d} P_H$. For $d = e$ (trivial coset) and for any coset where σ_d has a reduced expression with a left- and a right-descent in H, we have $P_H T_{\sigma_d} P_H \in (1+q)^{2s} H_q(S_n)$ by the $(1+T_i)T_{s_i} = q(1+T_i)$ pushing. The obstruction is cosets whose minimal-length representative has *no* descent in H on either side (the "diagonal" cosets), of which $\sigma = s_2$ at $n = 4$ is the smallest example.

This explains why the direct algebraic approach fails: the trace on any single module can still be divisible when summed over cosets, because spiky entries cancel — but this cancellation is a representation-theoretic fact (V^H has low dimension), not an algebra-level fact.

3.3 Why Lemma 4 does not extend to $n \geq 6$

The staircase word for $w_0 \in S_n$ reads

$$s_1 \mid s_2 s_1 \mid s_3 s_2 s_1 \mid s_4 s_3 s_2 s_1 \mid \cdots$$

For the identity $\Pi_q = AP_H B$ we need a contiguous sub-word whose indices, in some order, are precisely $\{1, 3, 5, \ldots\} \cap [1, n-1]$. At $n = 4$: positions $3, 4$ carry (s_1, s_3) — exact match. At $n = 5$: same positions, same match. At $n = 6$: $H = \langle s_1, s_3, s_5 \rangle$ needs indices $\{1, 3, 5\}$ but the staircase has no contiguous trio of commuting odd-indexed factors; the earliest occurrence of s_5 is position 11, and the three s_1, s_3, s_5 do not line up as a factor. Thus the Lemma 4 strategy does not extend.

4 Verification summary

- **Hoefsmit seminormal form.** Implemented per-irrep as $d \times d$ symbolic matrices over $\mathbb{Q}(q)$ with $d = \dim V_\lambda$. Checked: $T_i^2 = (q-1)T_i + q$, braid, far-commutativity for all irreps at $n = 3, 4, 5$. (`extract_n5_eigenvalues.py`)
- **Trace divisibility table.** All entries of Table 1 extracted via $\mathbb{Q}(q)$-matrix products then symbolic polynomial division. Ten irreps verified; every one passes.
- **Factorization $\Pi_q = AP_H B$.** Confirmed in the $n! \times n!$ length-lex regular representation: all 576 ($n = 4$) and 14,400 ($n = 5$) entries of $\Pi_q - AP_H B$ are zero. (`verify_APHB.py`)
- **Per-eigenvalue divisibility.** From characteristic polynomials of $\Pi_q|_{V_\lambda}$: for $\dim V_\lambda^H \geq 2$ cases $(4, 1)$ and $(3, 2)$ at $n = 5$, verified $(1+q)^s \mid S$ and $(1+q)^{2s} \mid P$ as above.
- **Counterexample for the algebra-level claim.** Confirmed that $P_H T_{s_2} P_H \notin (1+q)^2 H_q(S_4)$ by exhibiting specific coefficients. (`check_PH_yP_H_v2.py`)

5 What remains

1. **Integrality of $\mathrm{tr}_{V_\lambda^H}(Y|_{V_\lambda^H})$ (primary gap).** Proposition 6 reduces trace divisibility to proving that the trace on V_λ^H of BA (acting via $H_q(S_n)$) lies in $\mathbb{Z}[q]$, not merely in $(1+q)^{-?}\mathbb{Z}[q]$. This is equivalent to showing that the idempotent $e_H = P_H/(1+q)^s$ has an integer lift on V_λ — i.e., the projector onto V_λ^H is representable by an element of $\mathbb{Z}[q]_{(1+q)} \cdot H_q(S_n)$ with well-defined reduction $\mod (1+q)$. Over the DVR $\mathbb{Z}[q]_{(1+q)}$, P_H itself is nilpotent modulo $(1+q)$, so no naive lift exists; a more refined argument (parabolic Hecke algebra, base change to the residue field) is needed.
2. **Extension to $n \geq 6$.** The $\Pi_q = AP_H B$ factorization does not persist; the staircase word no longer admits the contiguous $\{1, 3, 5, \ldots\}$ subword. A different organization — e.g. the Soergel/EW-category $\Pi_q = BS(\mathbf{i}) = \bigoplus_w m_w B_w$ — would be insensitive to this accident of word position.
3. **Palindromic factor $p_\lambda(q)$.** The last column of Table 1 exhibits, after dividing by $q^{n(\lambda)}$, the proposed palindromic factors: for $n = 5$, $p_{(5)} = (1+q)^8$, $p_{(4,1)} = (1+q)^2(q^4+7q^3+17q^2+7q+1)$, $p_{(3,2)} = q^4+9q^3+19q^2+9q+1$, $p_{(3,1,1)} = q^2+4q+1$, $p_{(2,2,1)} = 1$. Each is palindromic and has non-negative integer coefficients. Interpreting p_λ as a graded trace or KL polynomial remains open.

Honest scope of this note

Theorem 3 is a *computational* theorem at $n \leq 5$, not a general-n structural proof. The partial algebraic skeleton (Lemma 4, Corollary 5, Proposition 6) reduces the question to an integrality statement we do not yet know how to prove, and in any case fails to extend to $n \geq 6$. The Soergel/BS($\mathbf{i}$) route — where $(1+q)^s$ is the Poincaré polynomial of the parabolic H and V^H-multiplicities are graded — remains the strongest candidate for a full proof. This note records the precise state of the proof effort as of 2026-04-23.

Code and data: the seminormal-form implementation and trace/divisibility verification scripts are in the scratch directory and reproduce Table 1 and every claim labelled "computational verification" above.

Table 1: Traces $\mathrm{tr}_{V_\lambda}(\Pi_q)$ for $n = 3, 4, 5$, computed via Hoefsmit seminormal form. $s = \lfloor n/2 \rfloor$. Rows with $V_\lambda^H = 0$ have $\Pi_q|_{V_\lambda} = 0$ by Lemma 2 and are omitted.

n	λ	$\dim V_\lambda$	$\dim V_\lambda^H$	$\mathrm{tr}_{V_\lambda}(\Pi_q)$	$\mathrm{tr}_{V_\lambda}(\Pi_q)/(1+q)^s$
3	(3)	1	1	$(1+q)^3$	$(1+q)^2$
3	$(2,1)$	2	1	$q(1+q)$	q
4	(4)	1	1	$(1+q)^6$	$(1+q)^4$
4	$(3,1)$	3	1	$q(1+q)^2(q^2+4q+1)$	$q(q^2+4q+1)$
4	$(2,2)$	2	1	$q^2(1+q)^2$	q^2
5	(5)	1	1	$(1+q)^{10}$	$(1+q)^8$
5	$(4,1)$	4	2	$q(1+q)^4(q^4+7q^3+17q^2+7q+1)$	$q(1+q)^2(q^4+7q^3+17q^2+7q+1)$
5	$(3,2)$	5	2	$q^2(1+q)^2(q^4+9q^3+19q^2+9q+1)$	$q^2(q^4+9q^3+19q^2+9q+1)$
5	$(3,1,1)$	6	1	$q^3(1+q)^2(q^2+4q+1)$	$q^3(q^2+4q+1)$
5	$(2,2,1)$	5	1	$q^4(1+q)^2$	q^4